

NR2DT-03: Step 3 — Design

Series: NR2DT | **Notebook:** 3 of 10 | **Created:** April 2026 | **Last Updated:** 04/14/2026

Overview

Goal of this step: lock in the target Dynatrace architecture before any artifact migrates. Bucket strategy, host group taxonomy, IAM model, and OpenPipeline routing are all decided here.

Procedural runbook — see **USFOODS-G.01** (host groups) and **USFOODS-G.02** (Grail buckets) for the recommended design patterns this step uses.

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Prerequisites

Requirement	Details
Audience	Migration lead + assigned engineer for this step

Requirement	Details
Completed	NR2DT-02 — Strategize
Format	Procedural step — use as a runbook; defer to NRLC for depth
NRLC deep dives	USFOODS-G.01 / G.02 (governance recommendations)

1. What You'll Produce

Artifact	Format	Purpose
bucket-design.md	Markdown	Bucket names, retention, pricing model
host-group-design.md	Markdown	Host group taxonomy and naming
iam-model.md	Markdown	Groups, policies, scopes
openpipeline-design.md	Markdown	Enrichment + routing + filter rules
terraform/	HCL	Phase 1 Terraform module skeleton

2. Bucket Design

Use the recommended pattern from **USFOODS-G.02 Grail Buckets** as your starting template:

Bucket	Retention	Pricing	Purpose
<org>_app_logs	30 days	Retain w/ Included Queries	Application logs
<org>_infra_logs	14 days	Retain w/ Included Queries	Infrastructure / K8s
<org>_security_logs	365 days	Usage-based	Audit trail / PCI
<org>_spans	14 days	Retain w/ Included Queries	Distributed traces
<org>_events	90 days	Retain w/ Included Queries	Platform events
<org>_bizevents	90 days	Retain w/ Included Queries	Business events

Use <org>_ as the prefix — it lets IAM policies scope with one STARTSWITH "<org>_" condition.

Document any deviations from this template (e.g., per-app buckets) and why.

3. Host Group Design

Per **USFOODS-G.01**, group hosts by ownership boundary, not by team alone. Common dimensions:

- Environment (`prod-app` , `nonprod-app` , `prod-infra`)
- Compliance scope (`payment-pci`)
- Workload tier (`prod-batch` , `prod-stream`)

Plan a host group per (environment × ownership) combination. Avoid a single all-hosts group — it breaks per-environment thresholds, IAM scoping, and automation safety.

4. IAM Model

Recommended starting groups:

Group	Bucket Access	Policy Condition
<code><org>-admins</code>	All	<code>STARTSWITH "<org>_"</code>
<code><org>-sre</code>	All operational	<code>bucket-name IN ('<org>_app_logs', '<org>_infra_logs', '<org>_events', '<org>_spans')</code>
<code><org>-app-eng</code>	App logs only	<code>bucket-name == '<org>_app_logs'</code>
<code><org>-readonly</code>	All except security	<code>STARTSWITH "<org>_" AND bucket-name != '<org>_security_logs'</code>

Always include an explicit DENY on `<org>_security_logs` for non-admin / non-audit groups.

5. OpenPipeline Layout

Routing rules in priority order (first match wins):

Priority	Matcher	Bucket
1	<code>matchesValue(log.source, "*audit*") OR loglevel == "CRITICAL"</code>	<code><org>_security_logs</code>
2	<code>in(k8s.namespace.name, {"kube-system", "kube-public", ...})</code>	<code><org>_infra_logs</code>
3	App namespace match	<code><org>_app_logs</code>
4	catch-all	<code>default_logs</code> (review weekly)

Enrichment rules for any NR workload identifier (replaces NR's tag/workload model):

```
matcher: contains(k8s.namespace.name, "prod")
field: environment
value: prod
```

Drop rules for cost optimization (decide which apply now vs. Wave 5):

- Drop health-check logs (`contains(content, "GET /health")`)
- Drop DEBUG logs in production
- Sample sidecar proxy logs

6. Step Exit Criteria

G3 — Architecture Locked

- [] `bucket-design.md` final; bucket names + retention + pricing model documented
- [] `host-group-design.md` final; taxonomy understood by SRE + service owners
- [] `iam-model.md` final; security stakeholder signed off
- [] `openpipeline-design.md` final; routing rules sequenced
- [] Phase 1 Terraform module skeleton committed (will be applied in Wave 0 of Step 5)

Next step: NR2DT-04 — Translate (NRQL→DQL run; manual triage).

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